

Software Development Fundamentals Lab – I [24B15CS111]

Lab 6

NOTE: All students must submit the programs (code with running output) of this assignment in doc format to their respective faculty of Lab. as per the file naming convention Enrollment No._Assignment No.

C172.1	Develop programs/logic for data types, expressions and conditional structure.
C172.2	Perform programs for arrays, strings and pointers

1. Write a C program to print hollow mirrored inverted right triangle star pattern series of n rows using for/while/do while loop.

Input rows: 5 Output

```
*****
****
***
**
*
```

2. Write a C program to count frequency of digits in an integer number given by user using for/while/do while loop.

Example

Input: Input any number: 116540

Output

```
Frequency of 0 = 1
Frequency of 1 = 2
Frequency of 2 = 0
Frequency of 3 = 0
Frequency of 4 = 1
Frequency of 5 = 1
Frequency of 6 = 1
Frequency of 7 = 0
Frequency of 8 = 0
Frequency of 9 = 0
```

3. Write a C program to find prime factors of a number given by user using for/ while/ do while loop.

Example

Input

Input any number: 10

Output

Prime factors of 10: 2, 5

4. Write a C program that will print the following pattern using for/while/do while loop.

```
      *
    * * *
  * * * * *
* * * * * *
* * * * * *
* * * * * *
* * * * * *
```

5. Write a C program that will print the following pattern using for/while/do while loop.



6. Write a C program to find the frequency of each element in an array.

Example

Input: array = {1, 2, 3, 5, 2, 9, 7, 3, 5}

Output:

1 occurs 1 time
2 occurs 2 times
3 occurs 2 times
5 occurs 2 times
7 occurs 1 time
9 occurs 1 time

7. Given an array, write a program to the left-rotate the elements of the array by one.

Example

Array Elements before rotating: 1 2 3 4 5 6 7

Array Elements after rotating: 2 3 4 5 6 7 1

8. A class has marks of N students in a subject stored in a 1D array. Write a C program to:

- Find the class average.
- Find the highest and second-highest marks.
- Find the number of students scoring above average.
- Find the longest consecutive sequence of students who scored above average.
- Display the positions of all students who obtained the highest marks.

Constraint: Do not sort the array.

9. An index is called an equilibrium index if the sum of elements on its left equals the sum of elements on its right.

Example

-7 1 5 2 -4 3 0

Output

Equilibrium Indices: 3, 6

Write a program to find **all equilibrium indices**.

10. An array contains numbers from 1 to N. Exactly one number is missing and another number occurs twice.

Example

1 3 4 5 3

Output

Missing = 2

Repeating = 3

Restrictions:

- Do not sort.
- Do not use another array.

11. Two arrays A and B of the same size are given. Determine whether B is a **circular rotation** of A.

Example

A = 1 2 3 4 5

B = 4 5 1 2 3

Output

Circular Rotation

12. A college stores marks of N students in M subjects in a 2D array.

Write a program to:

- Calculate total marks of each student.
- Assign a rank to every student **without sorting the matrix**.
- Find the subject in which the class performed best.
- Identify students who failed in more than one subject.
- Find the student with the highest improvement between two specified subjects.

13. Given a 9×9 matrix, determine whether it represents a valid Sudoku configuration.

Check:

- Every row contains 1–9 without repetition.
- Every column contains 1–9 without repetition.
- Every 3×3 sub-grid contains 1–9 without repetition.

Do not solve the Sudoku; only validate it.

14. Given an $N \times M$ matrix, print its elements in spiral order.

Example:

1 2 3

4 5 6

7 8 9

Output:

1 2 3 6 9 8 7 4 5

15. Given a matrix, if any element is 0, set its **entire row and column to zero**.

Example:

1 2 3

4 0 6

7 8 9

Output:

1 0 3

0 0 0

7 0 9

16. Use Virtual Lab portal for Arrays: <https://cse02-iiith.vlabs.ac.in/exp/arrays/index.html>